

Ex.no:3	DATA CLEANING

Aim:

To clean and preprocess the **California Housing Dataset** by identifying and handling missing values, removing duplicate and blank records, correcting data types, and preparing a consistent and reliable dataset for further housing data analysis.

Algorithm:

Check Missing Values

1. Start.
2. Import the Pandas library.
3. Load the California Housing Dataset into a DataFrame (df).
4. Display dataset information using df.info().
5. Check for missing values using df.isnull().
6. Count missing values in each column using df.isnull().sum().
7. Check missing values using df.isna().
8. Check non-missing values using df.notnull().
9. Display the missing-value results.

Drop Missing Values

1. Start.
2. Load the California Housing Dataset into a DataFrame (df).
3. Drop rows containing missing values using dropna().
4. Remove blank rows and duplicate records.
5. Display the cleaned California Housing Dataset.
6. Analyze housing records based on a selected **median house value** condition

Program:

```
import pandas as pd
```

```
# Load dataset
```

```
df = pd.read_csv("/content/housing prices.csv")
```

```
print("Dataset Loaded Successfully")

# Select important columns for demonstration
housing_df = df[
    [
        "longitude",
        "latitude",
        "median_income",
        "total_rooms",
        "median_house_value",
        "ocean_proximity"
    ]
].head(5)

print("\nOriginal DataFrame:")
display(housing_df)

print("\n1. First 5 rows:")
display(housing_df.head())

print("\n2. Last 2 rows:")
display(housing_df.tail(2))

print("\n3. Shape:")
print(housing_df.shape)

print("\n4. Column Names:")
print(housing_df.columns)

print("\n5. DataFrame Information:")
housing_df.info()

print("\n6. Statistical Summary:")
```

```
display(housing_df.describe())
```

```
print("\n7. Missing Values:")
```

```
print(df.isnull().sum())
```

```
print("\n8. Non-Missing Values:")
```

```
print(df.notnull().sum())
```

```
# Remove blank rows and duplicates
```

```
df = df.dropna()
```

```
df = df.drop_duplicates()
```

```
print("\n9. Cleaned Dataset:")
```

```
display(df.head())
```

```
print("\n10. Records with Median House Value greater than 300000:")
```

```
display(
```

```
    df[df["median_house_value"] > 300000][
```

```
    [
```

```
        "longitude",
```

```
        "latitude",
```

```
        "median_income",
```

```
        "total_rooms",
```

```
        "median_house_value",
```

```
        "ocean_proximity"
```

```
    ]
```

```
    ].head()
```

```
)
```

Output:

Original DataFrame:

	longitude	latitude	median_income	total_rooms	median_house_value	ocean_proximity
0	-122.23	37.88	8.3252	880.0	452600.0	NEAR BAY
1	-122.22	37.86	8.3014	7099.0	358500.0	NEAR BAY
2	-122.24	37.85	7.2574	1467.0	352100.0	NEAR BAY
3	-122.25	37.85	5.6431	1274.0	341300.0	NEAR BAY
4	-122.25	37.85	3.8462	1627.0	342200.0	NEAR BAY

1. First 5 rows:

	longitude	latitude	median_income	total_rooms	median_house_value	ocean_proximity
0	-122.23	37.88	8.3252	880.0	452600.0	NEAR BAY
1	-122.22	37.86	8.3014	7099.0	358500.0	NEAR BAY
2	-122.24	37.85	7.2574	1467.0	352100.0	NEAR BAY
3	-122.25	37.85	5.6431	1274.0	341300.0	NEAR BAY
4	-122.25	37.85	3.8462	1627.0	342200.0	NEAR BAY

2. Last 2 rows:

	longitude	latitude	median_income	total_rooms	median_house_value	ocean_proximity
3	-122.25	37.85	5.6431	1274.0	341300.0	NEAR BAY
4	-122.25	37.85	3.8462	1627.0	342200.0	NEAR BAY

3. Shape:

(5, 6)

4. Column Names:

```
Index(['longitude', 'latitude', 'median_income', 'total_rooms',  
      'median_house_value', 'ocean_proximity'],  
      dtype='object')
```

5. DataFrame Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 5 entries, 0 to 4

Data columns (total 6 columns):

#	Column	Non-Null Count	Dtype
0	longitude	5 non-null	float64
1	latitude	5 non-null	float64
2	median_income	5 non-null	float64
3	total_rooms	5 non-null	float64
4	median_house_value	5 non-null	float64
5	ocean_proximity	5 non-null	object

dtypes: float64(5), object(1)

memory usage: 372.0+ bytes

6. Statistical Summary:

	longitude	latitude	median_income	total_rooms	median_house_value
count	5.000000	5.000000	5.000000	5.000000	5.000000
mean	-122.238000	37.858000	6.674660	2469.400000	369340.000000
std	0.013038	0.013038	1.921878	2603.018114	47089.733488
min	-122.250000	37.850000	3.846200	880.000000	341300.000000

	longitude	latitude	median_income	total_rooms	median_house_value
25%	-122.250000	37.850000	5.643100	1274.000000	342200.000000
50%	-122.240000	37.850000	7.257400	1467.000000	352100.000000
75%	-122.230000	37.860000	8.301400	1627.000000	358500.000000
max	-122.220000	37.880000	8.325200	7099.000000	452600.000000

7. Missing Values:

```

longitude      0
latitude       0
housing_median_age  0
total_rooms     0
total_bedrooms  207
population     0
households     0
median_income   0
median_house_value  0
ocean_proximity  0
dtype: int64

```

8. Non-Missing Values:

```

longitude      20640
latitude       20640
housing_median_age  20640
total_rooms     20640
total_bedrooms  20433
population     20640
households     20640
median_income   20640
median_house_value  20640

```

ocean_proximity 20640

dtype: int64

9. Cleaned Dataset:

	longitud e	latitud e	housing_ median_a ge	total_ room s	total_b edroom s	popu lation	hous eholds	median _income	median_h ouse_valu e	ocean_ proximi ty
0	-122.23	37.88	41.0	880.0	129.0	322.0	126.0	8.3252	452600.0	NEAR BAY
1	-122.22	37.86	21.0	7099.0	1106.0	2401.0	1138.0	8.3014	358500.0	NEAR BAY
2	-122.24	37.85	52.0	1467.0	190.0	496.0	177.0	7.2574	352100.0	NEAR BAY
3	-122.25	37.85	52.0	1274.0	235.0	558.0	219.0	5.6431	341300.0	NEAR BAY
4	-122.25	37.85	52.0	1627.0	280.0	565.0	259.0	3.8462	342200.0	NEAR BAY

10. Records with Median House Value greater than 300000:

	longitud e	latitud e	median_incom e	total_room s	median_house_val ue	ocean_proximit y
0	-122.23	37.88	8.3252	880.0	452600.0	NEAR BAY
1	-122.22	37.86	8.3014	7099.0	358500.0	NEAR BAY
2	-122.24	37.85	7.2574	1467.0	352100.0	NEAR BAY
3	-122.25	37.85	5.6431	1274.0	341300.0	NEAR BAY

	longitud e	latitud e	median_incom e	total_room s	median_house_val ue	ocean_proximit y
4	-122.25	37.85	3.8462	1627.0	342200.0	NEAR BAY

Result: